NERAVU

Backend & AI Developer Handoff

Status and implementation plan | 3 November 2026 | Internal engineering document

Purpose. Onboard the incoming developer to take ownership of FastAPI, Supabase, scheme verification, Hybrid RAG, Sarvam LLM, eligibility/document engines, testing and deployment. The existing developer will build the React UI.

Evidence standard. “Confirmed” means supported by logs, database queries or API responses shared during development. “Planned” does not mean implemented. The live GitHub repository was not audited for this document.

1 | Executive snapshot

Area	What is confirmed	Open issue
API	FastAPI starts; /health and /docs HTTP 200; /api/chat HTTP 200 in English and Telugu	Scheme detail GET observed loading; debug database timeouts
Knowledge	PM-JAY draft record; ingestion reported 70 chunks; E5 and BGE loaded	Only one programme confirmed, not four; source verification incomplete
LLM	Sarvam 105B works; OpenAI-compatible API; response includes excerpts	Answer hallucination / ambiguity; no claim validation
Governance	programme_versions, verification_reviews, sync_jobs tables exist	No demonstrated authenticated publish or worker

2 | Product boundaries

• Primary users: cancer patients and caregivers in India; languages initially English, Telugu, Hindi.

• Core workflows: discover assistance, explain programme conditions, screen possible eligibility, prepare documents, guide official application steps, and track progress.

• Not a medical diagnostic tool or a funding authority. Human reviewers approve source facts; official agencies decide eligibility and applications.

• NGOs may only accept hospital referrals. Do not assume an NGO portal can accept direct patient submissions.

• Patient personal/medical data must not be added to the scheme knowledge base or logged in prompts.

3 | Confirmed technology

• Backend: Python FastAPI; SQLAlchemy connection to Supabase PostgreSQL.

• Embeddings: intfloat/multilingual-e5-small, 384 dimensions; passage:/query: prefixes; pgvector.

• Retrieval: hybrid lexical/vector search with reranking using BAAI/bge-reranker-v2-m3; inspect actual SQL and code before modifying.

• Generation: Sarvam Token Factory endpoint https://api.sarvam.ai/v2, model sarvam-105b, OpenAI SDK compatible.

• Frontend owner is implementing React UI; establish a versioned API contract now.

4 | Existing data and schema

Original tables: **organizations, programmes, programme_sources, eligibility_rules, document_requirements**. RAG table: **knowledge_chunks** with 384-dimension embedding and full-text tsvector. Governance tables added: **programme_versions, verification_reviews, sync_jobs**.

PM-JAY ID: `3e1a22bf-fed3-4d06-9aae-6a00bd9daabe`
Programme code: `CSV-F64FDF5C58ACCF8E2112`
Review status: **draft**; version 1; last_verified_at null. Other three schemes were discussed but not confirmed imported.

Important: The current RAG returns excerpts from draft questionnaire data. The patient-facing search must be changed to use only reviewed, published versions. Do not merely flip review_status without a source review.

5 | Current API contract

Route	Status	Notes
GET /health	Confirmed	Returns {"status":"ok"}
POST /api/chat	Confirmed	Input: question + language en/te/hi; response includes answer, sources, verification_status, disclaimer
GET /api/schemes	Code supplied	Confirm successful DB response
GET /api/schemes/{scheme_id}	Registered in Swagger	Observed hanging/loading; debug
Admin create/edit/review/publish	Not implemented	Requires authentication, authorization and audit

Integration contract for UI owner: Keep current /api/chat response stable. New fields (intent, answer_status, source URLs, evidence version) should be optional/additive until coordinated. Publish an OpenAPI JSON fixture and example errors. Never mark draft responses “verified.”

6 | The known correctness failure

An English answer interpreted source wording “70+ members” as a family of 70 people. The intended meaning appears to be members **aged 70 or older**; this must be checked against official policy. Telugu avoided that interpretation but included weakly relevant bank-account details. The failure originated in ambiguous stored text and insufficient factual validation, not an HTTP/API error.

• Correct the source-of-truth text after official evidence review.

• Regenerate affected chunks and embeddings; changing programmes.details alone leaves stale knowledge_chunks.

• Add structured facts for high-risk amounts, ages, deadlines, conditions and exceptions.

• Add claim-to-evidence checking and multilingual regression tests.

7 | Target architecture

Admin draft → evidence → review → publish → immutable version → sync jobs → published retrieval/rules/docs.

Patient question → intent router → relevant engine → verified evidence assembly → Sarvam → claim checks → answer/citations/unknown state.

Separate programme information from general cancer education. The latter needs curated authoritative medical sources, symptom/urgency routing, and no diagnosis. Eligibility engine returns possible match / mismatch / unknown, never official approval.

8 | Prioritized backend backlog with acceptance criteria

P0 / Repo and DB audit (0.5–1 h)

Clone, inspect code and requirements; verify migrations, .env paths, API routes and Supabase schema; reproduce GET hang.

Done when: GET /api/schemes and detail return bounded success/4xx/5xx; connection/statement timeouts set; no secrets printed.

P0 / Authentication and draft CRUD (2–4 h)

Implement admin-only create/read/update for programmes, sources and structured fields; input validation and authorization.

Done when: Unauthorized write rejected; editing creates auditable version history; existing PM-JAY data unchanged.

P0 / Evidence verification and publish (3–5 h)

Source URLs, issuing body, clause/page, effective date, reviewer and review timestamp; approve/reject; transactional publish.

Done when: Cannot publish missing/unreviewed critical claims; approved snapshot immutable; superseded version tracked.

P0 / Verified-only retrieval (2–4 h)

Index published versions with metadata; filter RAG to published/current; internal draft preview separated.

Done when: No draft chunk can be returned by patient API; changes trigger safe reindex and old-version retirement.

P1 / Sync worker (2–4 h)

Process sync_jobs for embeddings, rules, docs, guidance; idempotency, retries, error status and version guard.

Done when: Publishing creates jobs; successful run updates all views; failed job visible and retryable; no mixed versions.

P1 / Retrieval and LLM reliability (3–6 h)

Better chunking, hybrid search, rerank tests, intent routing, abstention and claim checks; limit irrelevant details.

Done when: Golden questions have grounded citations; age/amount facts consistent in en/te/hi; no unsupported invented claims in acceptance set.

P1 / Import and verify more schemes (variable)

Import three additional records as drafts; map 65-field template, check sources and clinical/hospital workflow.

Done when: Four schemes visible to admin only; each published record independently reviewed.

P1 / Rules, documents, journey (3–5 h)

Wire approved eligibility_rules, document_requirements and steps to APIs; unknown-state logic.

Done when: Rule explanations cite evidence; missing facts do not become false positives; checklists use current versions.

P2 / General cancer Q&A (3–6 h)

Separate medical education collection, safe routing, escalation for urgent symptoms, source review.

Done when: No diagnostic certainty; appropriate urgent-care guidance; tests for ambiguous/out-of-scope questions.

P2 / Deployment and security (3–6 h)

Auth, RLS/least privilege, rate limits, CORS, logging minimization, secrets, monitoring, backups, CI.

Done when: Staging smoke tests pass; no public admin routes; no patient-sensitive prompts in logs.

9 | Practical implementation design

Publish transaction: Lock programme; confirm reviewed evidence and allowed state; create immutable programme_versions snapshot; update published version pointer/status; insert sync_jobs with unique (programme_id, version, job_type); commit. Worker processes jobs with retries. Consider keeping previous published version active until all required indexes are ready, then atomically switch active version.

Indexing: Build chunks from approved version snapshot and verified source passages; include programme ID, source ID, page/section, version and validity dates. Generate normalized E5 embeddings. Upsert by stable content hash and version; retire stale chunks only after new index is ready. Existing table may require metadata columns or a new published index table.

Retrieval: Route question by intent; filter programme and approved version; lexical+vector search; rerank; deduplicate; impose relevance and coverage checks; pass bounded, labeled excerpts to Sarvam. Never allow source text to override system instructions.

Answer validator: Extract critical claims (money, age, deadlines, eligibility, application steps); compare against approved structured facts and cited passages; if unsupported, regenerate once or return “cannot confirm.” This improves safety but does not guarantee perfection.

Medical information: Curated independent collection; do not reuse financial-scheme chunks as medical evidence. Escalate urgent treatment-related symptoms and defer clinical decisions to oncology professionals.

10 | Testing checklist

• Unit: rule operators, unknown/contradictory facts, source-status guards, chunk version filtering, language validation.

• Integration: admin auth, CRUD, review, publish, job retries, verified-only patient queries, DB timeouts.

• Retrieval: per-scheme golden questions, missing-source abstention, contradictory circulars, stale version rejection, source citation correctness.

• LLM: 70+ age interpretation, family-vs-person caps, unrelated banking details, prompt injection in retrieved text, unsupported eligibility assertions.

• Multilingual: English, Telugu, Hindi same facts and numbers; assess answer language and citation preservation.

• Operational: large BGE model cold start/memory, Sarvam empty-content/finish-reason handling, API timeouts, provider outages, cost limits.

• Privacy: no secrets or personal health data in logs; least-privilege DB credentials; admin routes blocked from public access.

11 | Developer handoff: first day

Sequence	Deliverable	Share with UI owner
1	Run app locally; reproduce /health, /api/chat, /api/schemes; fix hanging DB read	Working base URL and API health status
2	Freeze v1 chat response; publish Swagger/OpenAPI and error schemas	Typed frontend request/response examples
3	Admin auth + draft CRUD + sources	Admin endpoint specification
4	Publish gate and versioning tests	Verified/draft status behavior
5	Verified-only RAG + reindex jobs	Citations, status, and latency expectations

12 | Collaboration and delivery

UI developer: Build patient/caregiver React screens with mock fixtures for unfinished APIs: onboarding/language choice, chat, scheme explorer/detail, citations/verification state, document checklist, application progress. Never present draft as verified. Backend developer owns API/data and admin auth.

Daily sync: 15 minutes to share endpoints, response schemas, DB migrations, blocking bugs and sample responses. Use small pull requests, branch protection, and code review. Do not share secrets in chat, tickets, screenshots or commits.

Timeline: Focused backend reliability and publishing MVP roughly 12–20 developer hours if current code is healthy; broader medical routing, security and production readiness typically require substantially more. Official verification of multiple schemes is an additional human review effort.

Repository note: This handoff and the companion README are generated artifacts. They have not been committed to the GitHub repository. Inspect and merge deliberately.